



SAVEETHA ENGINEERING COLLEGE

AUTONOMOUS

**NAAC
A
GRADE**

Accredited by
NBA
NATIONAL BOARD
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5 UG Programs

Affiliated to Anna University | Approved by AICTE
Saveetha Consortium of Euture Technologies
(SCOFT)

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE
B.Tech. Artificial Intelligence and Data Science (AI-DS)

28th Jan 2025



Agenda

- About the Curriculum - AI-DS
- Courses Offered
- Blended Learning
- Project

Guidelines

Guidelines for the upcoming Semester,

Attendance Requirements:

- In addition to the existing 75% attendance requirement, you must achieve 100% “learning attendance” in each course to be eligible for end-semester exams.
- “Learning attendance” is calculated by combining:
Physical class presence (excluding OD, ML, AB), plus Marks obtained from Attendance Bridging Assessments (ABA).
- Example Cases:
Case 1: A student has 85% overall attendance by having 55% physical presence, 30% OD, and 15% absence. To cover the 45% of physical absence, the student must take nine ABA exams.

[Important Instructions for Learners](#)

Guidelines for the upcoming Semester,

Prime and Slow Learners

- Prime students and slow learners do not need to enroll online. Their enrollments will be done manually.

Industry-Expert Courses

- 19AI513/CS3513 - Game Programming: This course will teach you how to design and develop 3D games using Maya and Unreal Engine.
- 19CS570/CS1570 - Entrepreneurship and Small Business Development: Learn the skills required to set up and manage a small startup business effectively.
- 19CS545/CS1545 - Open Source Operating System: Gain hands-on experience managing Red Hat Linux operating systems, preparing you for the Red Hat System Administration global certification.

[Important Instructions for Learners](#)

Guidelines for the upcoming Semester,

Study Hours (10:00 AM – 3:00 PM)

- Study hours are allocated whenever you have free time, with one weekday exception allowed based on your preference.
- Use this time for assignments, hackathon preparation, and other academic work.
- Attendance during study hours is mandatory to sit for end-semester exams.
- Missing two days of study hours results in suspension from CAMU.

Mentoring Sessions

- Held every Wednesday from 1:00 PM to 3:00 PM.
- Attendance is compulsory.
- Missing two mentoring sessions leads to suspension from CAMU.

[Important Instructions for Learners](#)

Guidelines for the upcoming Semester,

Courses for Third-Year Students (Targeting >10 LPA Jobs)

- Third-year students aiming for campus interviews with salary packages over 10 LPA should select “Graph Theory” and “System Design” during enrollment.
- Each is a professional elective worth 3 credits.
- Even if you have fulfilled your professional elective requirements, it's recommended to take these courses if you are pursuing offers above 10 LPA.
- **Prerequisite:** Completion of 18 modules is mandatory before choosing these courses.
- **19CS509/CS1509 - Graph Theory and Applications:** This course will teach you how to apply data structures and graph theory concepts to tackle complex coding challenges for competitive exams.
- **19IT505/CS2505 - Software Design Patterns:** You will learn to apply object-oriented design principles for developing complex system architectures.

[Important Instructions for Learners](#)

Guidelines for the upcoming Semester,

Instructor-Led Vivas for Technical Training

- After completing certain technical training modules, you must pass an instructor-led viva (minimum 75% to qualify for the next module).
- Vivas occur after Modules 2, 5, 8, 11, 14, 17, 20, and 23.
- If you fail a viva, you must repeat the last two modules before retaking it.

Blended Learning Courses

- Blended courses combine offline classes with online learning and are scheduled for two hours per week.
- These are labeled “Blended” in the timetable.
- To enroll in blended courses, you must have completed a certain number of technical training modules:
 - 1st Year: 6 modules
 - 2nd Year: 12 modules
 - 3rd Year: 18 modules

[Important Instructions for Learners](#)

Guidelines for the upcoming Semester,

Minor or Honors degree programs:

- Students enrolling in minor or honors degree programs are required to select all their courses during the online enrollment process. Separate manual course enrollments will not be available for minor or honors degrees.

Fullstack Courses:

- Full-stack courses are offered as separate components. Students must individually select courses for frontend development, server-side programming, and database management.
- Frontend:
19AI545-Modern Web Application Development (React)
- Server-side programming (Any One):
19AI553-Advanced Java Web Applications (Springboot)
19CS567-Cloud Native App Development (.Net Core)
- Database:
19CS404-Database Management System and its Applications

[Important Instructions for Learners](#)

Fully Flexible Choice Based Credit System (FFCBCS) - B. Tech - AI&DS



CATEGORY	REGULAR - Minimum credits to be earned	LATERAL ENTRY - Minimum Credits to be earned
	Credits	Credits
Humanities and Social Science Courses	12	3
Basic Science Courses	21	14
Engineering Science Courses	28	12
Professional Core Courses	56	48
Professional Elective Courses	16	16
Open Elective Courses	12	12
Employability Enhancement Courses	16	16
Total	161	121
Mandatory Courses	3	3

Courses Offered

HUMANITIES AND SOCIAL SCIENCE COURSES

(Minimum Credits to be earned: 12)

S. NO.	CODE	COURSE TITLE	HOURS PER WEEK			C	MIN CREDITS **
			L	T	P		
1	19EN104	Technical Writing	3	0	0	3	3
2	19EN105	Public Speaking	0	0	6	3	
3	19EN101	Communicative English	3	0	2	4	
4	19MS154	Basic Financial Accounting*	3	0	0	3	3
5	19MS156	Human Resource Management and Team Building*	3	0	0	3	3
6	19MS155	Stock Market and Company Operations*	3	0	0	3	3

BASIC SCIENCE COURSES

(Minimum Credits to be earned: 21)

S. NO.	CODE	COURSE TITLE	HOURS PER WEEK			C	MIN CREDITS **
			L	T	P		
1	19MA220	Mathematics for Artificial Intelligence	4	0	0	4	4
2	19MA221	Linear Algebra Laboratory	0	0	4	2	2
3	19CY205	Principles of Chemistry in Engineering*	3	0	2	4	4
4	19PH214	Physics for Quantum Computing*	3	0	0	3	3
5	19MA222	Probability And Queueing Models	3	0	2	4	4
6	19MA219	Transforms And Its Applications	3	1	0	4	4

*Courses exempted for lateral entry students ** Minimum credits to be earned

ENGINEERING SCIENCE COURSES

(Minimum Credits to be earned: 28)

S. NO.	CODE	COURSE TITLE	HOURS PER WEEK			C	MIN CREDITS**
			L	T	P		
1	19AI301	Python Programming	2	0	2	3	3
2	19AI302	Engineering Design And Modelling *	0	0	6	3	3
3	19AI303	Engineering Mechanics and Product Development *	2	0	2	3	3

4	19AI306	Object Oriented Programming using C++	2	0	2	3	3
5	19AI307	Object Oriented Programming using Java	2	0	2	3	
6	19AI308	Object Oriented Programming using C#	2	0	2	3	
7	19EE404	Digital Electronics*	3	0	2	4	4
8	19EE305	Basic Electrical, Electronics And Measurement Engineering*	2	0	2	3	3
9	19EE309	Programming Microcontrollers*	2	0	2	3	3
10	19AI304	Fundamentals of C Programming	2	0	2	3	3
11	19AI305	Advanced C Programming	2	0	2	3	3

*Courses exempted for lateral entry students ** Minimum credits to be earned

A. PROFESSIONAL CORE COURSES
(Minimum Credits to be earned: 56)

S. NO.	CODE	COURSE TITLE	HOURS PER WEEK			C	MIN CREDITS**
			L	T	P		
1	19AI414	Fundamentals of Web Application Development	3	0	4	5	5
2	19CS405	<u>Operating System</u> *	3	0	2	4	4
3	19CS406	<u>Computer Networks</u> *	3	0	2	4	4
4	19CS404	<u>Database Management System and Its Applications</u>	3	0	2	4	4
5	19MA211	<u>Statistics And Numerical Methods</u>	3	0	2	4	4

7	19AI408	Data Structures	2	0	2	3	3
8	19AI404	Analysis Of Algorithms	2	0	2	3	3
9	19AI405	Fundamentals Of Artificial Intelligence	2	0	2	3	3
10	19AI406	Digital Image Processing Techniques	2	0	2	3	3
11	19AI407	Parallel Computing Architecture	3	0	2	4	4
12	19AI403	Introduction to Data Science	2	0	2	3	3
13	19AI409	Applied Artificial Intelligence	2	0	2	3	3
14	19AI410	Introduction to Machine Learning	2	0	2	3	3
15	19AI411	Neural Networks	2	0	2	3	3
16	19AI412	Web Data Mining	3	0	2	4	4
17	19AI413	Deep Learning and its applications	2	0	2	3	3

*Courses exempted for lateral entry students ** Minimum credits to be earned

PROFESSIONAL ELECTIVE COURSES
(Minimum Credits to be earned: 16)

S. NO.	CODE	COURSE TITLE	HOURS PER WEEK			CREDITS
			L	T	P	
1	19CS407	Theory Of Computation	3	0	0	3
2	19CS408	Software Engineering	3	0	2	4
3	19AI521	Expert Systems	3	0	0	3
4	19AI509	Concepts Of Virtual And Augmented Reality	2	0	2	3
5	19AI505	Reinforcement Learning	2	0	2	3
6	19AI539	Mobile User Interface Development	2	0	2	3
7	19AI540	Programming Mobile Devices	2	0	2	3
8	19CS523	Information Retrieval	3	0	0	3

PROFESSIONAL ELECTIVE COURSES
(Minimum Credits to be earned: 16)

S. NO.	CODE	COURSE TITLE	HOURS PER WEEK			CREDITS
			L	T	P	
1	19CS417	Ethical Hacking Techniques	2	0	2	3
2	19CS420	Prototyping of IoT Systems	2	0	2	3
3	19CS542	Embedded Board Design	2	0	2	3
4	19AM508	Introduction to IoT	2	0	2	3
5	19AI513	Game Programming	2	0	4	4
6	19AM401	Time Series Analysis and Forecasting	3	0	2	4
7	19CS418	CYBER LAW AND COMPLIANCE	2	0	2	3
8	19AI547	Blockchain for Business	2	0	2	3

PROFESSIONAL ELECTIVE COURSES
(Minimum Credits to be earned: 16)

S. NO.	CODE	COURSE TITLE	HOURS PER WEEK			CREDITS
			L	T	P	
1	19CS409	Compiler Design	3	0	2	4
2	19CS305	Computer Architecture	3	0	0	3
3	19CS515	Blockchain Architecture and its Use Cases	3	0	0	3
4	19CS504	Software Project Management	3	0	0	3
5	19AI533	Introduction to Robotics	2	0	2	3
6	19AI541	<u>Cloud Computing</u>	3	0	0	3
7	19CS415	<u>Cryptography</u>	2	0	2	3
8	19AM509	<u>Industrial internet of things</u>	2	0	2	3

PROFESSIONAL ELECTIVE COURSES
(Minimum Credits to be earned: 16)

S. NO.	CODE	COURSE TITLE	HOURS PER WEEK			CREDITS
			L	T	P	
1	19CS416	Cloud Security				
2	19CS419	Digital Forensics and Digital Investigations				
3	19CS421	Edge Computing				
4	19CS422	Security for IoT Systems				
5	19AI555	Exploratory Data Analysis				
6	19AI516	Big Data Analytics				

F. OPEN ELECTIVE COURSES
(Credits to be earned: 12)

S. NO.	CODE	COURSE TITLE	CREDITS	REMARKS
1		Open Elective Courses	8	Courses offered by other departments
2	19OC601	Online Course 1	2	Approved Courses
3	19OC602	Online Course 2	2	

#Course will be offered by the Institution / Department in collaboration with industry.

EMPLOYABILITY ENHANCEMENT COURSES (Credits to be earned: 16)							
S. NO.	CODE	COURSE TITLE	HOURS PER WEEK			CREDI TS	Remarks
			L	T	P		
1	19AI701	Mini Project	0	0	2	1	III Year
2	19AI702	Project Work I	0	0	6	3	IV Year

3	19AI703	Project Work II	0	0	12	6	NIL
4	19EY701	Soft Skills	0	0	2	1	NIL
5	19EY702	Creative Skills for Communication	0	0	2	1	NIL
6	19EY703	System of Numerical and Logical Terminologies	0	0	2	1	NIL
7	19EY704	Advanced Quantitative and Logical Reasoning	0	0	2	1	NIL
8	19EY705	Employment Enhancement Skills	0	0	2	1	NIL
9	19EY706	Company-Specific Assessments for Employability	0	0	2	1	NIL
*Courses exempted for lateral entry students							

H. MANDATORY COURSES
(Credits to be earned: 3)

S. NO.	CODE	COURSE TITLE	CREDITS
1	19MC801	Professional Ethics	0
2	19MC802	Environmental Science	0
3	19MC803	Constitution of India	0
4	19MC804	Internship/Entrepreneurship/Consultancy	2*
5	19MC805	Inplant Training	1*
6	19MC807	NSS [#]	0
7	19MC808	NSO [#]	0
8	19MC809	YRC [#]	0

* Credits not included for CGPA. [#] Any one course to be taken

Combined Courses Title

S. No	Combined Course Title	Course title with course Code
1	19AI301C:Python and Linear Algebra	19AI201 - Mathematics for Artificial Intelligence 19AI202 - Linear Algebra Laboratory 19AI301 - Python Programming

Programming Course - Modules

Language	Module 1 - 6	Module 7-12	Module 13 - 18	Module 19 - 24
C Programming	Language basics Fundamentals of C Programming - 19AI304	Recursion Matrix operations Intro. Stack, Queue, Linked List, Hackerrank challenges. Advanced C Programming - 19AI305	Applications of stack, Queue, Tree, Graph Data Structures - 19AI408	
Python Programming	Language basics	Recursion Matrix operations Intro. Stack, Queue, Linked List, Hackerrank challenges.	Applications of stack, Queue, Tree, Graph	Dynamic programming, Greedy method, Back tracking, Divide and conquer, Approximation

Programming Course - Modules

Language	Module 1 - 6	Module 7-12	Module 13 - 18	Module 19 - 24
C++ Programming	Language basics, STL introduction.	Applications of stack, Queue, Tree, Graph		
Java Programming	Language basics	Threads, Exception handling, Streams, Collections		

19AI306 - Object Oriented Programming using C++

19AI307 - Object Oriented Programming using Java

Blended Learning

Blended Learning

- Our enhanced blended learning program, which offers equivalent online courses for the academic courses. This initiative is designed to provide our learners with a comprehensive educational experience by combining the best of online and in-person learning.
- Eligibility Criteria:
Prerequisite (Eligibility):
2nd Year Students: Completion of six technical training Modules
3rd Year Students: Completion of twelve technical training Modules

Contd..

Mode of Conduction:

Two hours of offline class per week (Group discussion and presentation)

Coursera online course access will be provided

CIA examination and end-semester exams

Blended Learning: Course List

19AI403-INTRODUCTION TO DATA SCIENCE

19AM508-INTRODUCTION TO IOT

19AI414-Fundamentals of Web Application Development

19AI407-Parallel Computing Architecture

19AI410-Introduction to Machine Learning

19CS411-Virtualization and cloud Computing

19AI413-Deep Learning and its Applications

19AI502-APPLIED NATURAL LANGUAGE PROCESSING

19CS567-Generative AI Application Development

19CS569-Applied Data Science

19CS404-Database Management System and Its Applications

Project

Project

19AI702 - Project Work II

Learners can form a team of maximum of three members for Project Work 1.

Teams can be composed of members from the same mentor or from different mentors. There is no need to change your mentor for the project work.

Q & A

THANK YOU...